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# Characterisation of the acute hypoxic response using breathing variability parameters: A pilot study in humans

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ABSTRACT

*Purpose:* We aimed to investigate respiratory rate variability (RRV) and tidal volume ($V_t$) variability during exposure to normobaric hypoxia (i.e., reduction in the fraction of inspired oxygen - $FiO_2$), and the association of the changes in RRV and $V_t$ variability with the changes in pulse oxygen saturation ($SpO_2$).
*Methods:* Thirty healthy human participants (15 females) were exposed to: (1) 15-min normoxia, (2) 10-min hypoxia simulating 2200 m, (3) 10-min hypoxia simulating 4000 m, (4) 10-min hypoxia simulating 5000 m, (5) 15-min recovery in normoxia. Linear regression modelling was applied with $SpO_2$ (dependent variable) and the changes in RRV and $V_t$ variability (independent variables), controlling for $FiO_2$, age, sex, changes in heart rate (HR), changes in HR variability (HRV), and changes in minute ventilation ($V_E$).
*Results:* When modelling breathing parameter variability as root-mean-square standard deviation (RMSSD), a significant independent association of the changes in RRV with the changes in $SpO_2$ was found (B = −4.3e-04, 95% CI = −8.3e-04/−2.1e-05, p = 0.04). The changes in $V_t$ variability showed no significant association with the changes in $SpO_2$ (B = −1.6, 95% CI = −5.5/2.4, p = 0.42). When modelling parameters variability as SD, a significant independent association of the changes in RRV with the changes in $SpO_2$ was found (B = −8.2e-04, 95% CI = −1.5e-03/−9.4e-05, p = 0.03). The changes in $V_t$ variability showed no significant association with the changes in $SpO_2$ (B=1.4, 95% CI = −5.8/8.6, p = 0.69).
*Conclusion:* Higher RRV is independently associated with lower $SpO_2$ during acute hypoxic exposure, while $V_t$ variability parameters are not. Therefore, RRV may be a potentially interesting parameter to characterize individual responses to acute hypoxia.

## 1. Introduction

Exposure to acute hypoxia influences the human body homeostasis and induces several physiological responses. The decreased partial pressure of inspired oxygen (due to reduced atmospheric pressure at altitude or due to reduced fraction of inspired oxygen, $FiO_2$) reduces partial pressure of arterial oxygen ($PaO_2$), which, in turn, is detected by the highly sensitive peripheral chemoreceptors in the carotid body. Consequently, minute ventilation ($V_E$) increases, improving alveolar ventilation and restoring, at least in part, $PaO_2$ levels. The increased $V_E$ reduces partial pressure of arterial carbon dioxide ($PaCO_2$), triggering the peripheral chemoreceptors to modulate the degree of hyperventilation (Steinback and Poulin, 2007). This regulatory process is referred to as acute hypoxic ventilatory response (AHVR). At the same time, sympathetic activation leads to an increased heart rate (HR) and pulmonary vasoconstriction to ensure appropriate ventilation-perfusion matching (Heistad and Abboud, 1980).

The AHVR is primarily driven by an increase in tidal volume ($V_t$),

*Abbreviations:* AHVR, acute hypoxic ventilatory response; $f_B$, Breathing frequency; $FiO_2$, fraction of inspired oxygen; $PaCO_2$, partial pressure of arterial carbon dioxide; $PaO_2$, partial pressure of arterial oxygen; RRV, Respiratory rate variability; RMSSD, root-mean-square standard deviation; SDBB, standard deviation of breath-to-breath intervals; SDNN, standard deviation of normal-to-normal interbeat intervals; $SpO_2$, pulse oxygen saturation; VE, Minute ventilation; Vt, Tidal volume; HR, Heart rate; HRV, Heart rate variability.

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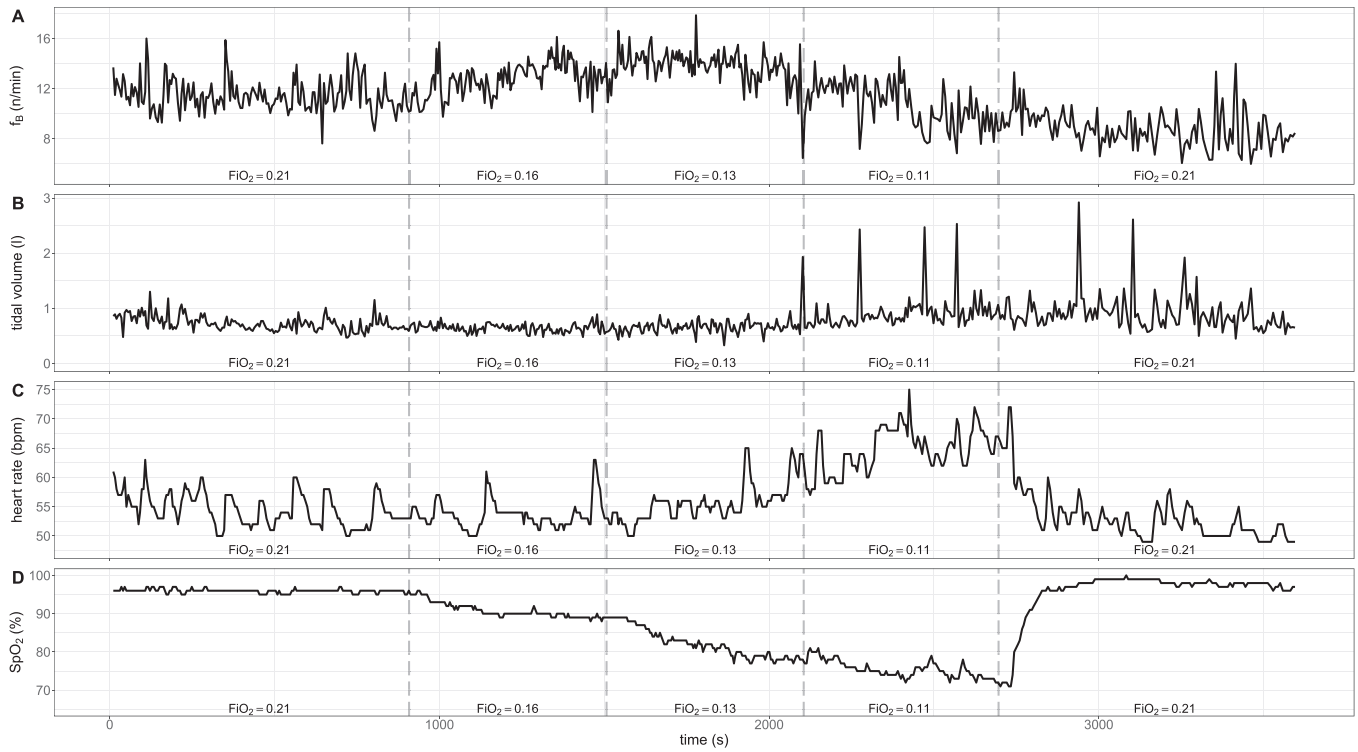


**Fig. 1.** Representative raw recordings of a single study participant throughout the whole recording. $f_B$, breathing frequency; $SpO_2$, pulse oxygen saturation; $FiO_2$, fraction of inhaled oxygen.

rather than breathing frequency ($f_B$) (Bender et al., 1987; Mateika et al., 2004). This is important as a primary increase in $f_B$ would elevate physiological dead space ventilation and correct $PaO_2$ less efficiently. In line with this reasoning, studies found that individuals exhibiting a disproportionate increase in $f_B$ as AHVR are more susceptible to acute mountain sickness (Nespoulet et al., 2012). As the responses to hypoxia show large inter-individual differences, monitoring pulse oxygen saturation ($SpO_2$) is not sufficient to characterize the complex physiological responses and should be complemented with other parameters such as the ventilatory response. Researchers proposed to characterise the AHVR with various parameters. They range from the quantification of increase in $V_t$ (Jubran and Tobin, 2000; Waggener et al., 1984), and the quantification of contribution of $f_B$ to the increase in $V_E$ (Mateika et al., 2004), to indexes expressing the amplitude of $V_E$ increase relative to the $SpO_2$ reduction (Duffin, 2007). Finally, beyond the increase in HR induced by acute hypoxic exposure, HR variability (HRV) was proposed to complement information on AHVR from a cardiorespiratory perspective. However, evidence on its association with AHVR remains inconclusive (Oliveira et al., 2017; Povea et al., 2005; Taralov et al., 2018; Yamamoto et al., 1996). In particular, it is known that the breathing pattern is prone to modify HRV, which may complicate its interpretation under hypoxic conditions (Hermand et al., 2019; van den Aardweg and Karemaker, 1991).

The ventilatory effect of hypoxic exposure is characterized not only by an increase in $V_E$ due to the stimulation of the respiratory centres, but also by the induction of some degree of respiratory instability. Earlier work quantified this phenomenon using the periodicity of $V_E$ oscillations (Hermand et al., 2021; Hermand et al., 2015, 2019); or the variability in $SpO_2$ (Costello et al., 2020; Jiang et al., 2021). Adding to this approach, the instability of the respiratory system under hypoxic conditions can be characterized by the variability of $V_t$ and by the variability of the time intervals between successive breaths (respiratory rate variability; RRV). RRV is commonly used in critical care settings for the prediction of extubation failure (Seely et al., 2014) and chronic disease decompensation (Forleo et al., 2015). In neonates, RRV is used as a marker for stability of, possibly premature, breathing control, providing information regarding the risk of apnoeic events (Gallacher et al., 2016). Also, RRV has shown independent associations with blood pressure levels in hypertensive women, possibly predicting sleep apnoea (Anderson et al., 2008).

Thus, we aimed to investigate RRV and $V_t$ variability during hypoxic exposure with decreasing $FiO_2$, and the association of the changes in RRV and $V_t$ variability with the changes in $SpO_2$. We hypothesized that RRV and $V_t$ variability would increase with lower $SpO_2$, representing an increased respiratory instability.

## 2. Material and methods

### 2.1. Study design and participants

We performed an open-label pilot study exposing healthy participants to a single session of normobaric hypoxia at rest. All participants were residing at low altitude (i.e., <1000 m above sea level) and not regularly exposed to hypoxic conditions during the preceding six months. We included participants above 18 years of age and without any diagnosed clinical conditions. Approval was obtained from the local ethics committee (CPP Sud-Est V) and the study adhered to the Declaration of Helsinki. All participants provided written informed consent.

### 2.2. Experimental procedure

Participants arrived at the laboratory not having ingested heavy meals and caffeine for at least 4 h. In addition, they refrained from heavy exercise for at least 24 h prior to the experiment.

The experimental procedure took place in a quiet, temperature-controlled room. Via a facemask with a two-way valve, participants were connected to a hypoxia generator with a built-in pneumotachograph and $SpO_2$ sampling (Altitrainer, SMTEC, Switzerland). The pneumotachograph was calibrated with a 3 L syringe before every experiment. The exposure protocol consisted of five phases: (1) 15-min

**Table 1**
Participant characteristics (n = 30).

| Variable | |
|---|---|
| Age, y | 33 (12) |
| Sex, female/male (%) | 15/15 (50/50) |
| Weight, kg | 72.6 (18.9) |
| Height, cm | 176 (12) |
| Resting $SpO_2$, % | 98 (2) |
| Resting heart rate, bpm | 68 (16) |

Data are mean (SD) or n (%). SpO2, pulse oxygen saturation; bpm, beats per minute.

baseline phase under room air ($FiO_2 = 0.21$), (2) 10-min hypoxic phase simulating an altitude of 2200 m ($FiO_2 = 0.16$), (3) 10-min hypoxic phase simulating an altitude of 4000 m ($FiO_2 = 0.13$), (4) 10-min hypoxic phase simulating an altitude of 5000 m ($FiO_2 = 0.11$), (5) 15-min recovery phase under room air ($FiO_2 = 0.21$). During the protocol, the participants rested in a comfortable chair and did not speak to avoid artefacts in the recordings. Data sampling ran continuously.

HR was continuously monitored with a portable monitor (GT9X, ActiGraph, USA) and a heart rate belt (H7, Polar, Finland).

Dyspnoea ratings were collected during the last minute of each phase by pointing on a modified Borg Scale, ranging from 0 (no dyspnoea at all) to 10 (highest possible level of dyspnoea).

A representative raw signal plot of a single participant is presented in Fig. 1 with marks identifying the hypoxic phases.

### 2.3. Data analysis

All results are shown as mean (SD) unless stated otherwise. Variable distributions were determined visually using quantile-quantile plots. A two-sided p-value $< 0.05$ was considered statistically significant.

We applied linear regression modelling to determine the association between the changes in $SpO_2$ (dependent variable) and the changes in RRV and $V_t$ variability (independent variables). The model was controlled for $FiO_2$, age, sex, changes in HRV, changes in HR, and changes in $V_E$. Modelling was applied twice, expressing RRV and $V_t$ variability, and HRV i) as root-mean-square standard deviation (RMSSD), and ii) as standard deviation of breath-to-breath intervals (SDBB), standard deviation of $V_t$, and standard deviation of normal-to-normal interbeat intervals (SDNN), respectively.

The RRV Poincaré plot parameters SD1, an index for short-term variability, and SD2, an index for long-term variability, were obtained. Thereby, SD1 expressing perpendicular spread of breath-to-breath intervals to the line of identity, and SD2 expressing the spread along the line of identity.

Ventilatory, $SpO_2$, and HR data were extracted from the devices with dedicated software at a sampling rate of 100 Hz. Ventilatory data was processed in Python 3.10.4 for Windows (Python Software Foundation, www.python.org). The RRV parameter calculations were done with the NeuroKit2 package (Makowski et al., 2021). All other data processing and further calculations were done in R 4.2.0 for Windows (R Core Team 2022, R Foundation for Statistical Computing, Vienna, Austria), the HRV calculations were done with the RHRV package (García Martínez et al., 2017). RRV, $V_t$ variability, and HRV are reported as time-domain parameters from the last 5 min of each phase.

## 3. Results

Thirty participants (15 females and 15 males) completed this pilot study. Detailed sample characteristics are listed in Table 1. Characteristics did not differ among sex and responses were not different between sexes. On average, we included 35 (22), 33 (21), 33 (21), 31 (20), and 33 (21) breaths per participant for the analysis of the baseline, 2200 m, 4000 m, 5000 m, and recovery phase, respectively.

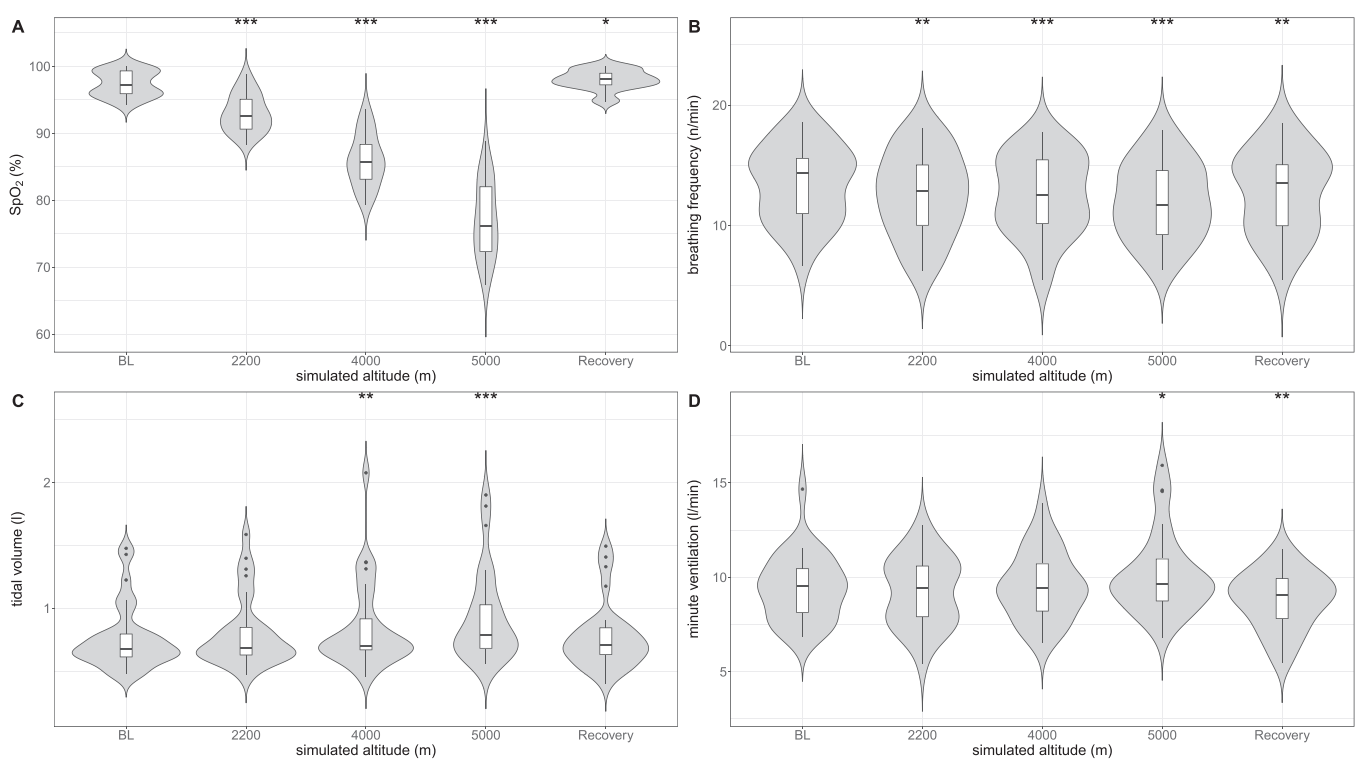


**Fig. 2.** Box- and violin plots (representing the distribution of data points) showing data for $SpO_2$ (A), breathing frequency (B), tidal volume (C), and minute ventilation (D) during the experimental phases. P-values refer to comparisons with the baseline value (BL). $SpO_2$, pulse oxygen saturation; BL, baseline. *, significantly different from Normoxia (baseline) (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

**Table 2**
Arterial oxygen saturation and ventilatory parameters during the experimental phases.

| Phase | $SpO_2$ (%) | $f_B$ (n/min) | $V_t$ (l) | $V_E$ (l/min) | Dyspnoea |
|---|---|---|---|---|---|
| Normoxia (baseline) | 97 (2) | 14 (3) | 0.7 (0.3) | 9.5 (3.1) | 0 (0) |
| Hypoxia ($FiO_2 = 0.16$) | 93 (3) *** | 13 (3) ** | 0.8 (0.3) | 9.5 (3.0) | 0 (0) |
| Hypoxia ($FiO_2 = 0.13$) | 86 (4) *** | 13 (3) ** | 0.8 (0.4) ** | 9.6 (3.1) | 1 (1) |
| Hypoxia ($FiO_2 = 0.11$) | 78 (6) *** | 13 (3) ** | 0.9 (0.4) *** | 10.2 (4.0) * | 1 (1) |
| Normoxia (recovery) | 98 (2) * | 13 (4) * | 0.7 (0.3) | 8.9 (3.1) ** | 0 (0) |

Data are mean (SD). SpO2, pulse oxygen saturation; fB, breathing frequency; Vt, tidal volume; VE, minute ventilation; FiO2, fraction of inhaled oxygen; Dyspnoea was measured on a scale from 0 (no dyspnoea) to 10 (maximal dyspnoea); * , significantly different from Normoxia (baseline) (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

### 3.1. Ventilatory and cardiorespiratory parameters

$SpO_2$, $f_B$, $V_t$, and $V_E$ are displayed in Fig. 2 for each experimental phase and absolute values are provided in Table 2. $SpO_2$ and $f_B$ were significantly reduced in all hypoxic phases compared to baseline. Breath-by-breath $f_B$ for all participants for each experimental phase are presented in Fig. 3. In Fig. 3, variability gets less towards the end of each phase. This is due to the fact that a decreasing number of participants is represented (i.e., only the ones with higher $f_B$ and thus more breaths). $V_t$ was significantly elevated at 4000 and 5000 m compared to baseline. $V_E$ was significantly elevated at 5000 m and significantly reduced at recovery compared to baseline. HR was significantly elevated in all hypoxic phases and significantly reduced at recovery compared to baseline (Table 3). Dyspnoea did not increase significantly at any phase of the protocol (Table 2).

### 3.2. HRV, RRV, and $V_t$ variability

HRV RMSSD was significantly reduced during all hypoxic phases compared to baseline. HRV SDNN was significantly reduced at 4000 m compared to baseline (Table 3), also when corrected for mean HR (using the formula $\frac{SDNN}{e^{-HR/58.8}}$ from Monfredi et al., 2014, results not shown).

RRV and $V_t$ variability are displayed in Fig. 4 for each experimental phase and absolute values are listed in Table 4. RRV RMSSD and SDBB were significantly elevated at 4000 m, 5000 m, and recovery compared to baseline. $V_t$ RMSSD and SD were significantly elevated at 5000 m compared to baseline.

RRV RMSSD and $V_t$ RMSSD ($r = 0.55$, $p < 0.001$), and RRV SDBB and $V_t$ SD ($r = 0.59$, $p < 0.001$) were significantly correlated.

### 3.3. Regression modelling

Linear regression modelling with RMSSD parameters of RRV, $V_t$ variability, and HRV showed a significant independent association of the changes in RRV with the changes in $SpO_2$ ($B = -4.3e\text{-}04$, 95% Confidence Interval [95% CI] = −8.3e-04/−2.1e-05, $p = 0.04$). The changes in $V_t$ variability showed no significant association with the changes in $SpO_2$ ($B = -1.6$, 95% CI = −5.5/2.4, $p = 0.42$). Adjusted $R^2$ of the

**Table 3**
Heart rate and its variability parameters during the experimental phases.

| Phase | HR (bpm) | HRV RMSSD (ms) | HRV SDNN (ms) |
|---|---|---|---|
| Normoxia (baseline) | 64 (13) | 78.2 (47.8) | 87.1 (35.9) |
| Hypoxia ($FiO_2 = 0.16$) | 65 (12) * | 69.4 (43.9) * | 82.4 (33.3) |
| Hypoxia ($FiO_2 = 0.13$) | 68 (13) *** | 60.7 (44.1) ** | 76.9 (32.2) * |
| Hypoxia ($FiO_2 = 0.11$) | 73 (14) *** | 58.5 (37.6) ** | 81.2 (38.2) |
| Normoxia (recovery) | 59 (10) *** | 83.9 (43.7) | 119.7 (44.1) *** |

Data are mean (SD). HR, heart rate; HRV, heart rate variability; RMSSD, root-mean-square standard deviation; SDNN, standard deviation of normal-to-normal interbeat intervals; FiO2, fraction of inhaled oxygen; * , significantly different from Normoxia (baseline) (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

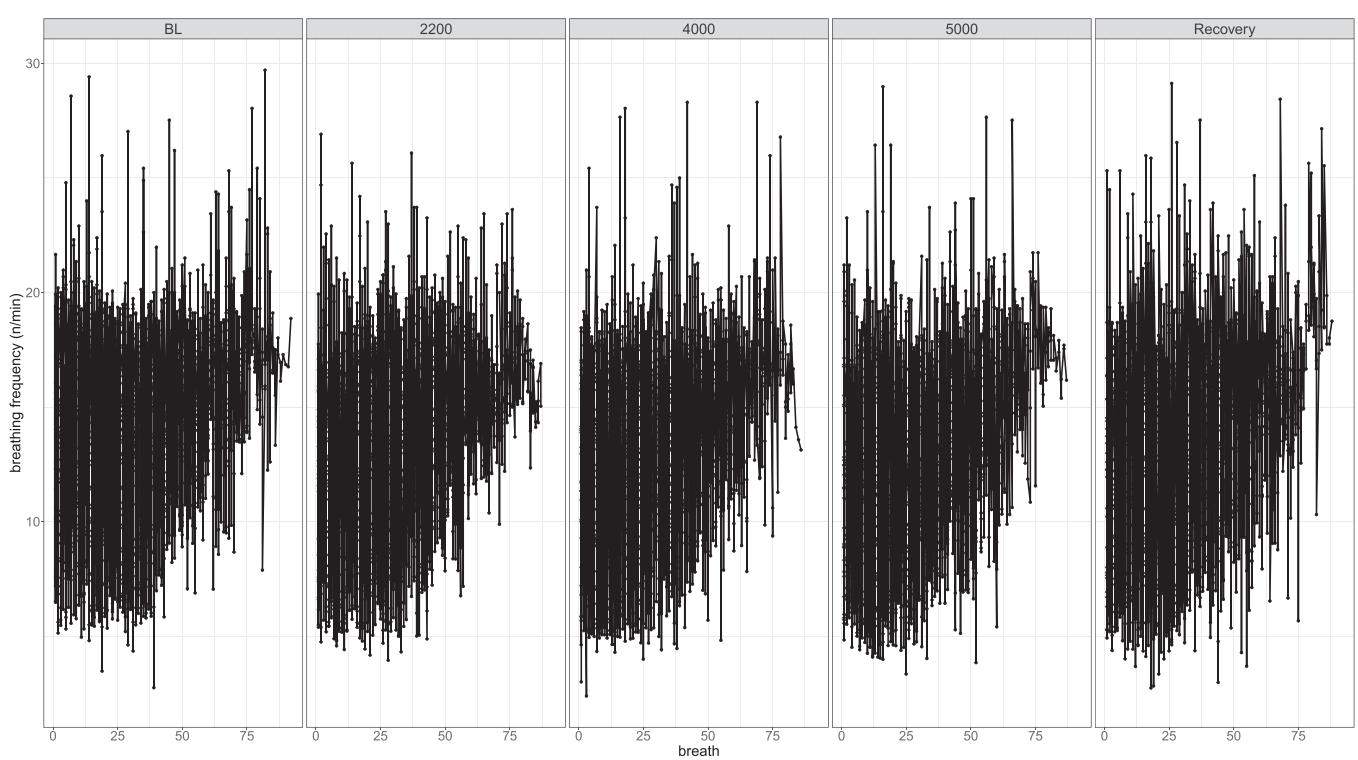


**Fig. 3.** Breath-by-breath breathing frequency across the study sample at all hypoxic phases. The x-axis shows breath position. BL, baseline.

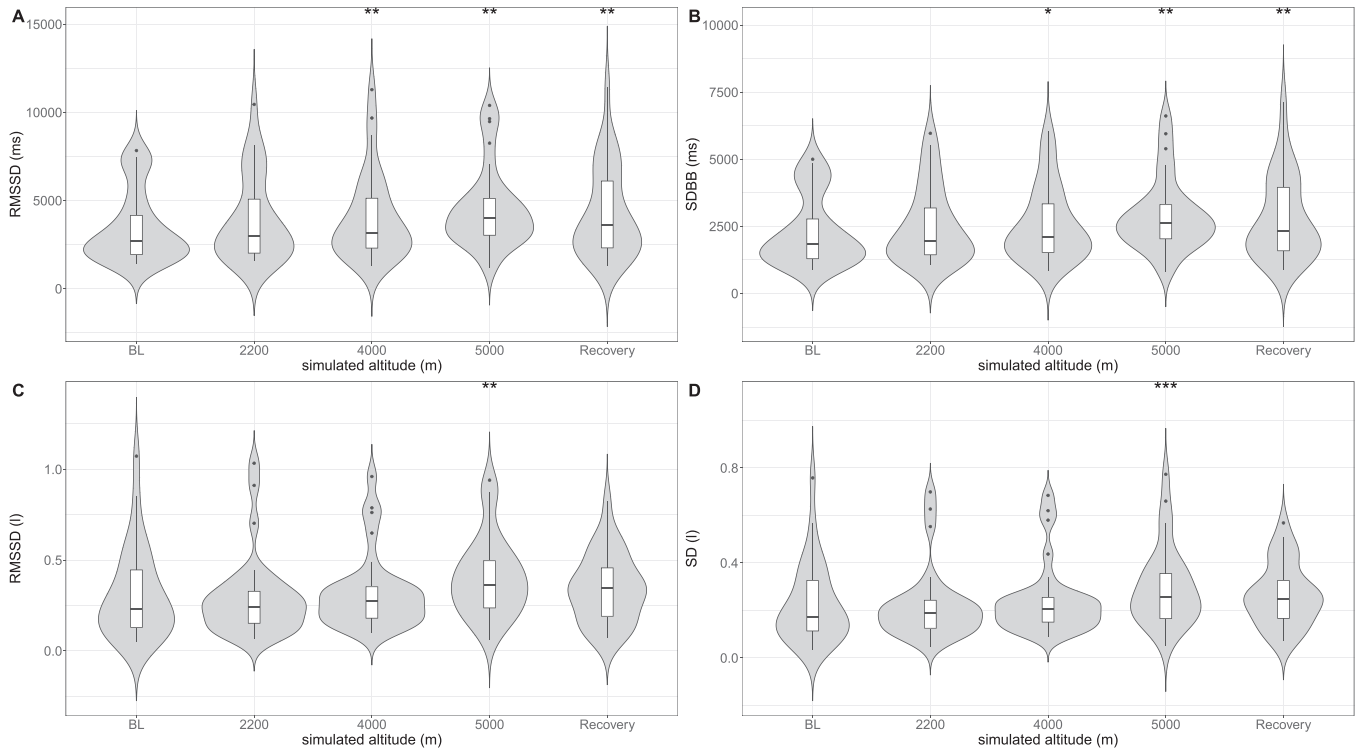


**Fig. 4.** Box- and violin plots (representing the distribution of data points) showing data for RRV RMSSD (A), SDBB (B), and $V_t$ RMSSD (C), SD (D) during the experimental phases. P-values refer to comparisons with the baseline value (BL). RRV, respiratory rate variability; $V_t$, tidal volume; RMSSD, root-mean-square standard deviation; SDBB: standard deviation of breath-to-breath intervals, SD, standard deviation; BL, baseline. * , significantly different from Normoxia (baseline) (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

**Table 4**
Tidal volume and respiratory rate variability parameters during the experimental phases.

| Phase | Vt RMSSD (l) | Vt SD (l) | RRV RMSSD (ms) | RRV SDBB (ms) | RRV SD1 (ms) | RRV SD2 (ms) |
|---|---|---|---|---|---|---|
| Normoxia (baseline) | 0.3 (0.2) | 0.2 (0.2) | 3465 (2033) | 2303 (1288) | 2475 (1460) | 2100 (1122) |
| Hypoxia ($FiO_2$ = 0.16) | 0.3 (0.2) | 0.2 (0.2) | 3908 (2407) | 2525 (1403) | 2795 (1734) | 2185 (1047) |
| Hypoxia ($FiO_2$ = 0.13) | 0.3 (0.2) | 0.2 (0.2) | 4054 (2486) ** | 2569 (1375) * | 2900 (1794) | 2140 (881) |
| Hypoxia ($FiO_2$ = 0.11) | 0.4 (0.2) ** | 0.3 (0.2) *** | 4582 (2320) ** | 2924 (1367) ** | 3278 (1674) | 2497 (1031) |
| Normoxia (recovery) | 0.3 (0.2) | 0.3 (0.1) | 4264 (2543) ** | 2794 (1554) ** | 3051 (1834) | 2481 (1271) |

Data are mean (SD). Vt, tidal volume; RRV, respiratory rate variability; RMSSD, root-mean-square standard deviation; SD, standard deviation; SDBB, standard deviation of breath-to-breath intervals; SD1, short-term variability; SD2, long-term variability; FiO2, fraction of inhaled oxygen; * , significantly different from Normoxia (baseline) (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

model was 87%. All coefficients are listed in Table 5.

Linear regression modelling with SDBB, SD, and SDNN parameters of RRV, $V_t$ variability, and HRV showed a significant independent association of the changes in RRV with the changes in $SpO_2$ (B = −8.2e-04, 95% CI = −1.5e-03/−9.4e-05, p = 0.03). The changes in $V_t$ variability showed no significant association with the changes in $SpO_2$ (B = 1.4, 95% CI = −5.8/8.6, p = 0.69). Adjusted $R^2$ of the model was 87%. All coefficients are listed in Table 6.

**Table 5**
Coefficients of the linear regression models with RMSSD on changes in $SpO_2$ to characterize physiological parameter variability.

| Independent Variable | Coefficient | 95% confidence interval | p-value |
|---|---|---|---|
| Δ RMSSD RRV | -4.3e-04 | -8.3e-04/− 2.1e-05 | 0.04 |
| Δ RMSSD $V_t$ | -1.6 | -5.5/8.6 | 0.42 |
| Δ RMSSD HRV | 2.5e-05 | -2.0e-02/2.0e-02 | 1.00 |
| Hypoxia ($FiO_2$ = 0.16) | -4.0 | -5.6/− 2.5 | < 0.001 |
| Hypoxia ($FiO_2$ = 0.13) | -10.8 | -12.5/− 9.2 | < 0.001 |
| Hypoxia ($FiO_2$ = 0.11) | -19.1 | -21.0/− 17.3 | < 0.001 |
| Normoxia (recovery) | 0.7 | -1.0/2.5 | 0.40 |
| Age | -1.6e-02 | -5.8e-02/2.6e-02 | 0.46 |
| Sex, male | -0.1 | -1.2/0.9 | 0.80 |
| Δ heart rate | -9.2e-02 | -0.2/0.0e-02 | 0.14 |
| Δ $V_E$ | 0.5 | -0.1/1.0 | 0.10 |

Dependent variable was the change in SpO2. SpO2, pulse oxygen saturation; RMSSD, root-mean-square standard deviation; Δ, Difference to baseline; RRV, respiratory rate variability; Vt, tidal volume; HRV, heart rate variability; FiO2, fraction of inhaled oxygen; VE, minute ventilation.

## 4. Discussion

This study focused on the significance of RRV and $V_t$ variability parameters under hypoxic conditions and its association with decreasing $FiO_2$ and $SpO_2$. Both RRV and $V_t$ variability parameters (RMSSD and SDBB/SD) increased with the severity of hypoxia. The main finding of this work is the independent association of the changes in RRV with the reduction in $SpO_2$, induced by stepwise decreases in $FiO_2$, suggesting that RRV might be a useful marker of hypoxic response.

The exposure to hypoxia as an environmental stressor activates a complex cascade of chemical sensing, ultimately leading to coupled regulation of the respiratory and the cardiovascular system (Khoo, 2000). Our findings suggest that breathing variability parameters provide distinct and valuable information on acute hypoxic responses

**Table 6**
Coefficients of the linear regression models with standard deviation changes in $SpO_2$ to characterize physiological parameter variability.

| Independent Variable | Coefficient | 95% confidence interval | p-value |
|---|---|---|---|
| Δ SDBB RRV | -8.18e04 | -1.5e-03/− 9.4e-05 | 0.03 |
| Δ SD $V_t$ | 1.4 | -5.8/8.6 | 0.69 |
| Δ SDNN HRV | -1.7e-03 | -2.2e-02/2.0e-02 | 0.88 |
| Hypoxia ($FiO_2 = 0.16$) | -4.0 | -5.6/− 2.5 | < 0.001 |
| Hypoxia ($FiO_2 = 0.13$) | -10.9 | -12.5/− 9.3 | < 0.001 |
| Hypoxia ($FiO_2 = 0.11$) | -19.2 | -21.1/− 17.4 | < 0.001 |
| Normoxia (recovery) | 0.7 | -1.2/2.5 | 0.49 |
| Age | -1.8e-02 | -6.0e-02/2.3e-02 | 0.37 |
| Sex, male | 1.1e-02 | -1.1/1.0 | 0.98 |
| Δ heart rate | -9.7e-02 | -0.2/2.7e-02 | 0.12 |
| Δ $V_E$ | 0.3 | -0.3/0.9 | 0.27 |

Dependent variable was the change in SpO2. SpO2, pulse oxygen saturation; SDBB: standard deviation of breath-to-breath intervals; SD, standard deviation; SDNN, standard deviation of normal-to-normal interbeat intervals; Δ, Difference to baseline; RRV, respiratory rate variability; Vt, tidal volume; HRV, heart rate variability; FiO2, fraction of inhaled oxygen; VE, minute ventilation.

compared to changes in $V_E$ and HR parameters. RRV seems to be a sensitive marker in detecting changes in arterial oxygenation. Its association with oxygen desaturation is significant even though the first two phases of our hypoxic protocol induced only slight decreases in $SpO_2$. Physiological adjustments of breathing were also visible when considering $V_t$, the observed augmentations being statistically significant from baseline at 4000 and 5000 m. Interestingly, the sensitivity of $V_t$ variability parameters appeared less clear, with only $V_t$ RMSSD being significantly augmented in severe hypoxia (i.e., when simulating 5000 m). RRV and $V_t$ variability were moderately correlated, and thus, it seems important to consider both dimensions of the respiratory cycle (i.e., volume and time dimensions). Building on our findings, future research might determine the strength of contribution and predictive ability of these dimensions during hypoxic exposure and eventually come up with a combined marker characterizing individual hypoxic responses. A recent study in mice found that $V_t$ variability and RRV might be controlled by differing pathways (MacMillan and Evans, 2022). The study found that breathing instability (i.e., RRV) increased in AMPK-$\alpha_1/\alpha_2$ knockout mice, while the response in $V_E$ remained unaffected (MacMillan and Evans, 2022). These findings combined with the association of RRV with $SpO_2$ in the present study suggest that RRV might provide useful information on breathing stability and AHVR.

Exposure to hypercapnia would potentiate the loop gain of the breathing control centres and hypercapnic hypoxia might induce even more pronounced fluctuations in RRV (Jubran et al., 1997; Steinback and Poulin, 2007). This hypothesis has to be kept in mind when translating the application of RRV parameters into clinical practice, especially in populations with respiratory diseases. For instance, individuals with chronic obstructive pulmonary disease or obstructive sleep apnoea exhibit higher RRV than healthy matched controls in a normoxic environment (Loveridge et al., 1984; Pal et al., 2022).

High interindividual variability was observed in hypoxia when considering $V_t$, $SpO_2$, and HR. This was especially true with higher levels of hypoxia (i.e., simulating 4000 and 5000 m). This finding could be associated with differences in acute mountain sickness susceptibility among the participants (Nespoulet et al., 2012). Since subjects were exposed to less than one hour of hypoxia within the present study, the hypothetical predictive value of RRV regarding acute mountain sickness could not be tested and should be considered in future studies.

Our findings provide some interesting insights regarding the recovery after a hypoxic exposure that would need further clarification. During the normoxic recovery phase, $SpO_2$ restored back to equal and even higher levels as compared to the baseline phase. In contrast, RRV recovered only partially and stayed elevated. Earlier work on the AHVR to poikilocapnic hypoxia showed, in line with our work, a lowered $f_B$ and similar $V_t$ as compared to baseline during recovery (Bender et al., 1987; Steinback and Poulin, 2007). We can only speculate on the underlying reasons for the persisting elevation of RRV, which may show that the hypoxic dose experienced caused some long-lasting instability in breathing. It remains to be elucidated if the observed augmentation in RRV persists for more than 15 min after termination of the hypoxic exposure and if there is an association with the magnitude of hypoxia experienced. In addition, the RRV might be depending on the mean $f_B$ during the phase, similar to HRV (Monfredi et al., 2014). Thus, modelling studies applying correction factors to RRV are needed. Finally, $f_B$ showed a slight reduction during the hypoxic phases in our work and only partially restored during the recovery phase. This might be a consequence of the corresponding changes in $V_t$, i.e. larger breathing amplitude leading to lower $f_B$.

Our findings add an additional perspective to recent exciting findings in variability-based analysis of $V_t$ and $SpO_2$ (Costello et al., 2020; Hermand et al., 2019; Jiang et al., 2021). Modelling and big data analysis may corroborate the existing findings and possibly come up with more refined and corrected analysis approaches. Ultimately shedding light on the measurement of respiratory instability in normobaric hypoxia. Last, the concept of RRV analysis is not profoundly established yet and we therefore believe that results regarding indexed parameters (i.e., SD1 and SD2) should be interpreted with caution.

This study has some limitations. First, we used convenience sampling and a highly standardised experimental setting. Since this is a pilot study, we could not perform a sample size calculation based on previous results. Also, we used a highly standardised setting, avoiding distraction and artefacts to a maximum and therefore reducing confounding factors. Hence, we encourage the confirmation of our results in distinct samples and real-life applications (e.g., at altitude or in patients). Second, we applied a stepwise increasing hypoxic protocol. An abrupt exposure to low $FiO_2$ might trigger slightly different responses in ventilatory variability than we observed. Last, our study cannot determine if RRV might distinguish between a well-compensated and a deleterious hypoxic stimulus. Insights into this domain would require exposures triggering acute mountain sickness symptoms.

In conclusion, higher RRV is independently associated with lower $SpO_2$ during acute hypoxic exposure, while $V_t$ variability parameters are not. Therefore, RRV may be a potentially interesting parameter to characterize individual hypoxic responses. It might be used to determine the hypoxic doses likely to induce efficient and positive responses (within hypoxic training or conditioning programs for instance) or conversely some level of intolerance (e.g., acute mountain sickness). Further studies should investigate ventilatory variability under hypoxic conditions coupled to different situations (i.e., rest, exercise, and sleep).

## Competing interests

The authors declare no conflicts of interest.

## Data Availability

Data will be made available on request.

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